Skills Review

Exercise No. 17

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Course/Section: CPE4B **Date: Oct. 7, 2026**

Machine Problem:

Write a program that will play the Paper, Rocks, Scissors game. To make sure that each player will not see his opponent's inputted letter, use function 07h to accept the single character input (P, R, or S) of the player. (For the purpose of monitoring the correctness of input character, function 01h may be used during the program development and testing. Just make sure that the final program will use function 07h instead of 01h). Make the program more flexible by allowing the players to enter either lowercase or uppercase letters. The program must repeat its execution until the players decide to terminate. Use as many functions and designs that have already been discussed before. The sample basic output below may be used as a guide. Save the program using Exer17.asm as the filename.

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.data

board   DB '               +-------------------------------------------------+',13,10
        DB '               |                                                 |',13,10
        DB '               |           PAPER, ROCK, SCISSORS GAME            |',13,10
        DB '               |    [P] Paper     [R] Rock     [S] Scissors      |',13,10
        DB '               |                                                 |',13,10
        DB '               |    Player 1:             Player 2:              |',13,10
        DB '               |                                                 |',13,10
        DB '               |                                                 |',13,10
        DB '               |                                                 |',13,10
        DB '               |                                                 |',13,10
        DB '               +-------------------------------------------------+',13,10,'$'

turn1   DB '        Player 1, press P, R or S (hidden)        $'
turn2   DB '        Player 2, press P, R or S (hidden)        $'
again   DB '    Play again? Press Y to continue, N to exit    $'

ready   DB 'READY   $'         ; shown while the move is still hidden
paper   DB 'Paper   $'
rock    DB 'Rock    $'
sciss   DB 'Scissors$'

p1win   DB '                 Player 1 wins!                $'
p2win   DB '                 Player 2 wins!                $'
draw    DB '         DRAW! Nobody wins this round.         $'
prmsg   DB '               Paper covers Rock!              $'
rsmsg   DB '             Rock crushes Scissors!            $'
```

```
spmsg   DB '              Scissors cut Paper!              $'
same    DB '       Both players picked the same move.      $'

.code
start:
        mov ax, @data          ; initialize the data segment
        mov ds, ax

restart:
        ;** clear the screen **
        mov ah, 06h            ; Function 06h - scroll up / clear window
        mov al, 00h            ; AL = 0 - clear the whole window
        mov bh, 07h            ; attribute - white on black
        mov cx, 0000h          ; upper left corner (row 0, col 0)
        mov dx, 184Fh          ; lower right corner (row 24, col 79)
        int 10h

        ;** display the game board **
        mov ah, 02h            ; Function 02h - set cursor position
        mov bh, 00h            ; page 0
        mov dx, 0300h          ; row 3, col 0
        int 10h

        mov ah, 09h            ; Function 09h - display string
        mov dx, offset board
        int 21h

        ;** player 1 turn **
        mov ah, 02h
        mov bh, 00h
        mov dx, 0F0Fh          ; row 15, col 15 (status line)
        int 10h

        mov ah, 09h
        mov dx, offset turn1   ; "Player 1, press P, R or S"
        int 21h

        call getmove           ; get a valid move (hidden)
        mov bl, al             ; BL = player 1 move

        mov ah, 02h
        mov bh, 00h
        mov dx, 081Eh          ; row 8, col 30 (after "Player 1:")
        int 10h

        mov ah, 09h
        mov dx, offset ready   ; show READY, not the move
        int 21h

        ;** player 2 turn **
        mov ah, 02h
        mov bh, 00h
```

```
        mov dx, 0F0Fh          ; row 15, col 15 (status line)
        int 10h

        mov ah, 09h
        mov dx, offset turn2   ; "Player 2, press P, R or S"
        int 21h

        call getmove           ; get a valid move (hidden)
        mov cl, al             ; CL = player 2 move

        ;** reveal both moves **
        mov ah, 02h
        mov bh, 00h
        mov dx, 081Eh          ; row 8, col 30 (player 1 move)
        int 10h

        mov al, bl             ; AL = player 1 move
        call getname           ; DX = offset of the move name
        mov ah, 09h
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dx, 0834h          ; row 8, col 52 (player 2 move)
        int 10h

        mov al, cl             ; AL = player 2 move
        call getname           ; DX = offset of the move name
        mov ah, 09h
        int 21h

        ;** decide the winner **
        cmp bl, cl             ; same move?
        jne notsame            ; if not, check who wins
        jmp tie                ; same move - it is a draw (too far for je)

notsame:
        cmp bl, 'P'            ; player 1 picked Paper?
        je p1paper
        cmp bl, 'R'            ; player 1 picked Rock?
        je p1rock
        jmp p1sciss            ; otherwise player 1 picked Scissors

p1paper:
        cmp cl, 'R'            ; Paper vs Rock
        je w1pr                ; player 1 wins
        jmp w2sp               ; Paper vs Scissors - player 2 wins

p1rock:
        cmp cl, 'S'            ; Rock vs Scissors
        je w1rs                ; player 1 wins
        jmp w2pr               ; Rock vs Paper - player 2 wins
```

```
p1sciss:
        cmp cl, 'P'             ; Scissors vs Paper
        je w1sp                 ; player 1 wins
        jmp w2rs                ; Scissors vs Rock - player 2 wins

        ;** display the result **
w1pr:   call line1              ; cursor at the winner line
        mov ah, 09h
        mov dx, offset p1win    ; "Player 1 wins!"
        int 21h
        call line2              ; cursor at the reason line
        mov ah, 09h
        mov dx, offset prmsg    ; "Paper covers Rock!"
        int 21h
        jmp askagain

w1rs:   call line1
        mov ah, 09h
        mov dx, offset p1win    ; "Player 1 wins!"
        int 21h
        call line2
        mov ah, 09h
        mov dx, offset rsmsg    ; "Rock crushes Scissors!"
        int 21h
        jmp askagain

w1sp:   call line1
        mov ah, 09h
        mov dx, offset p1win    ; "Player 1 wins!"
        int 21h
        call line2
        mov ah, 09h
        mov dx, offset spmsg    ; "Scissors cut Paper!"
        int 21h
        jmp askagain

w2pr:   call line1
        mov ah, 09h
        mov dx, offset p2win    ; "Player 2 wins!"
        int 21h
        call line2
        mov ah, 09h
        mov dx, offset prmsg    ; "Paper covers Rock!"
        int 21h
        jmp askagain

w2rs:   call line1
        mov ah, 09h
        mov dx, offset p2win    ; "Player 2 wins!"
        int 21h
        call line2
```

```
        mov ah, 09h
        mov dx, offset rsmsg   ; "Rock crushes Scissors!"
        int 21h
        jmp askagain

w2sp:   call line1
        mov ah, 09h
        mov dx, offset p2win   ; "Player 2 wins!"
        int 21h
        call line2
        mov ah, 09h
        mov dx, offset spmsg   ; "Scissors cut Paper!"
        int 21h
        jmp askagain

tie:    call line1
        mov ah, 09h
        mov dx, offset draw    ; "DRAW! Nobody wins this round."
        int 21h
        call line2
        mov ah, 09h
        mov dx, offset same    ; "Both players picked the same move."
        int 21h

        ;** play again? **
askagain:
        mov ah, 02h
        mov bh, 00h
        mov dx, 0F0Fh          ; row 15, col 15 (status line)
        int 10h

        mov ah, 09h
        mov dx, offset again   ; "Play again? Y / N"
        int 21h

getkey: mov ah, 07h            ; Function 07h - read a key without echo
        int 21h

        cmp al, 'Y'            ; is it Y?
        je yes                 ; if so, play again
        cmp al, 'y'            ; also accept lowercase y
        je yes
        cmp al, 'N'            ; is it N?
        je finish              ; if so, end the game
        cmp al, 'n'            ; also accept lowercase n
        je finish
        jmp getkey             ; any other key - wait for Y or N

yes:    jmp restart            ; restart is too far for je, so use jmp

finish:
        mov ah, 02h            ; move the cursor below the board
```

```
        mov bh, 00h             ; so the DOS prompt will not overlap
        mov dx, 1100h           ; row 17, col 0
        int 10h

        mov ah, 4Ch             ; terminate program
        int 21h

;** getmove - wait for a valid move, returns AL = 'P', 'R' or 'S' **
getmove:
        mov ah, 07h             ; Function 07h - read a key without echo
        int 21h                 ; AL = key pressed (not shown on screen)

        cmp al, 'P'             ; Paper?
        je valid
        cmp al, 'p'             ; lowercase p?
        je topaper
        cmp al, 'R'             ; Rock?
        je valid
        cmp al, 'r'             ; lowercase r?
        je torock
        cmp al, 'S'             ; Scissors?
        je valid
        cmp al, 's'             ; lowercase s?
        je toscissors
        jmp getmove             ; any other key - wait again

topaper:
        mov al, 'P'             ; store as uppercase P
        ret
torock: mov al, 'R'             ; store as uppercase R
        ret
toscissors:
        mov al, 'S'             ; store as uppercase S
        ret
valid:  ret                     ; valid move in AL

;** getname - AL = move, returns DX = offset of its name **
getname:
        mov dx, offset paper
        cmp al, 'P'             ; Paper?
        je done
        mov dx, offset rock
        cmp al, 'R'             ; Rock?
        je done
        mov dx, offset sciss    ; otherwise Scissors
done:   ret

;** line1 - cursor at the winner line **
line1:  mov ah, 02h
        mov bh, 00h
        mov dx, 0A10h           ; row 10, col 16
        int 10h
```

```
        ret

;** line2 - cursor at the reason line **
line2:  mov ah, 02h
        mov bh, 00h
        mov dx, 0B10h          ; row 11, col 16
        int 10h
        ret

end start
```

Output:

